

A Single-pass Noise Covariance Estimation Algorithm in Multiple-model Adaptive Kalman Filtering

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Abstract—Estimation of unknown noise covariances in a Kalman filter is a problem of significant practical interest in a wide array of industrial applications. The reliable algorithms for their estimation are scarce, and the necessary and sufficient conditions for the identifiability of the covariances were in dispute until very recently. In this paper, we present a single-pass stochastic gradient descent (SGD) algorithm for noise covariance estimation for use in multiple-model adaptive Kalman filters. The multiple-model estimation algorithm assumes that the system obeys one of a finite number of models, and each model has its own dynamics. The overall estimate of the multiple-model approach is a convex combination of the estimates from the filters running in parallel based on the individual models, where the weights of the convex combination are the concomitant posterior model probabilities. Here, the identifiability conditions of the multiple-model approach depend on each candidate model since the corresponding Kalman filters are non-interacting. The proposed streaming algorithm reads measurement data exactly once and has an acceptable root mean square error (RMSE) of state estimates. The proposed estimating algorithm is implemented based on its single-pass nature, recursive fading memory estimation of the sample cross-correlations of the innovations, and the root mean square propagation (RMSprop) accelerated SGD algorithm. Our approach enforces the structural assumptions on unknown noise covariances, and ensures the symmetry and positive definiteness of the estimated covariance matrices. The proposed method is applicable to non-stationary systems where the noise covariances continuously change. The validation of the proposed method on several numerical examples demonstrates its accuracy and filter consistency.

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1. INTRODUCTION

The Kalman filter (KF) [1] is an optimal minimum mean square error (MMSE) state estimator under the Gaussian assumption with known mean and covariance for discrete-time linear dynamic systems. It is also the best linear state estimator if the noises entering into the system are non-Gaussian with known first and second moments, i.e., mean and covariance. As a result of numerous research studies,

the Kalman filter has received substantial interest in many industrial applications, such as diagnosis, robotics, signal processing, navigation, and target tracking, to name a few [2, 3]. However, the statistics of noise processes are mostly unknown or partially known in many practical situations.

In [4], we presented the necessary and sufficient conditions and then presented a *batch* optimization algorithm for estimating unknown noise covariances. The noise covariance identifiability conditions involve the minimization of the sum of the normalized temporal cross-correlations of the innovations. With the Gaussian assumption, the innovation sequence of an optimal Kalman filter is orthogonal, implying that the innovations are independent of each other, and are strictly white [3].

In [5], we proposed an enhanced covariance estimation method based on a *sequential* mini-batch stochastic gradient descent (SGD) algorithm that required multiple passes through the data as well as a method to detect changes in noise covariances when applied to non-stationary systems. In this paper, we propose a *single-pass* adaptive Kalman filtering algorithm for non-stationary systems, wherein the process and measurement noise covariances continuously change, and the system behavior can stem from a member of a known subset of models.

Prior work

The key to estimating the noise covariances is a relationship linking the covariance of the state estimation error and of the innovations in any suboptimal filter. This relationship serves as a fundamental building block for the correlation-based approaches. Pioneering contributions using this approach were made by [6–9].

Sarkka and Nummenmaa [10] proposed a joint recursive noise-adaptive Kalman filter for linear state space models using variational Bayesian approximations. However, the variational methods generally require tuning hyper-parameters to converge to the correct covariance parameters and do not account for changes in process noise.

In [5], we improved the computational efficiency and accuracy of the batch estimation algorithm in [4] by a sequential mini-batch estimation method with adaptive step size rules. When updating the filter gain, we applied sequential fading memory mini-batch estimates of the innovation correlations. We also applied dynamic convergence thresholds to enhance the computational efficiency of the sequential estimation algorithm. The application of this method to non-stationary systems requires a change-point detection algorithm described in [11] to extract the time points of abrupt changes in unknown noise covariances based on the innovation sequence.

The multiple-model estimation algorithm assumes that the system obeys one of a finite number of models, and each model has its own non-switching dynamics [3, 12]. The overall estimate of multiple-model algorithm is a convex combination of the estimates from multiple parallel filters, each based on the individual models, where the weights of convex combination are the concomitant posterior model probabilities.

In this paper, we seek to overcome these limitations by developing a streaming algorithm that reads measurements exactly once, thus making it work for real-time estimation. We also extend the approach to multiple models.

Contribution and organization of the paper

We present a single-pass, sequential mini-batch noise covariance estimation algorithm as an extension of the work in [5, 13] for non-stationary systems. Our proposed method enables the estimates of the measurement and process noise covariances without the use of a change-point detection algorithm. The noise covariances are assumed to change continuously. More significantly, the structure of the dynamic model is unknown, but is known to belong to a known subset of models. We validate the proposed method on several non-stationary and multiple-model system test cases.

The paper is organized as follows. In Section 2, we provide an overview of the sequential mini-batch stochastic gradient descent (SGD) method in the multiple-model method, including identifiability conditions for estimating the unknown noise covariances in each individual model, the fading memory filter-based innovation correlation estimation, and the SGD update of the Kalman gain. Section 3 shows evidence that our method can track unknown noise covariances in non-stationary systems, as well as systems exhibiting dynamics from a known subset. Lastly, we conclude the paper and discuss potential avenues in Section 4.

2. SEQUENTIAL MINI-BATCH GRADIENT DESCENT ESTIMATION ALGORITHM IN MULTIPLE-MODEL APPROACH

The multiple-model approach is used in parallel and the system obeys one of a finite number of fixed models. Let us formally consider the discrete-time linear dynamic system can assume one of J models, $j = 1, 2, \dots, J$, given by

$$x(k+1) = F^j x(k) + \Gamma^j v^j(k) \quad (1)$$

$$z(k) = H^j x(k) + w^j(k) \quad (2)$$

where $x(k)$ is the n_x -dimensional state vector, $z(k)$ is the n_z -dimensional measurement vector, and j is the candidate model. F^j and H^j are the $n_x \times n_x$ state transition matrix and the $n_z \times n_x$ measurement matrix of the system, respectively, and Γ^j is the noise gain matrix. Here, the process noise $v^j(k)$ and the measurement noise $w^j(k)$ are the sequences of zero-mean white Gaussian noises with *unknown* process noise covariance $Q^j(k)$ and *unknown* measurement noise covariance $R^j(k)$, respectively. We assume that two noise covariance sequences and the initial state error are mutually uncorrelated. Note that $Q^j(k)$ and $R^j(k)$ continuously change in this paper.

Given $Q^j(k)$ and $R^j(k)$, the multiple-model adaptive Kalman filter involves the consecutive processes of prediction and update given by [1, 3, 14–16]

$$\hat{x}^j(k+1|k) = F^j \hat{x}^j(k|k) \quad (3)$$

$$\nu^j(k+1) = z(k+1) - H^j \hat{x}^j(k+1|k) \quad (4)$$

$$\hat{x}^j(k+1|k+1) = \hat{x}^j(k+1|k) + W^j(k+1) \nu^j(k+1) \quad (5)$$

$$P^j(k+1|k) = F^j P^j(k|k) F^{j'} + \Gamma^j Q^j(k) \Gamma^{j'} \quad (6)$$

$$S^j(k+1) = H^j P^j(k+1|k) H^{j'} + R^j(k) \quad (7)$$

$$W^j(k+1) = P^j(k+1|k) H^{j'} S^j(k+1)^{-1} \quad (8)$$

$$P^j(k+1|k+1) = (I_{n_x} - W^j(k+1) H^j) P^j(k+1|k) (I_{n_x} - W^j(k+1) H^j)' + W^j(k+1) R^j(k) W^{j'}(k+1)' \quad (9)$$

$$\Lambda^j(k) = \frac{1}{\sqrt{|2\pi S^j(k)|}} \exp\left(-\frac{1}{2} \nu^j(k)' S^j(k)^{-1} \nu^j(k)\right) \quad (10)$$

$$p^j(k) = \frac{\Lambda^j(k) p^j(k-1)}{\sum_{l=1}^J \Lambda^l(k) p^l(k-1)} \quad (11)$$

The Kalman filter predicts the next state estimate at time index $(k+1)$, given the observations up to time index k in (3) and the concomitant predicted state estimation error covariance in (6), using model-specific system dynamics, the updated state error covariance $P^j(k|k)$ at time index k and the process noise covariance, $Q^j(k)$. The updated state estimate at time $(k+1)$ in (5) incorporates the measurement at time $(k+1)$ via the Kalman gain matrix in (8), which depends on the innovation covariance $S^j(k+1)$ (which in turn depends on the measurement noise covariance $R^j(k)$, and the predicted state error covariance $P^j(k+1|k)$). The updated state error covariance $P^j(k+1|k+1)$ is computed via (9). This corresponds to Joseph form in [3, 17], which guarantees that the updated state covariance matrix will remain positive definite.

The mode likelihood function $\Lambda^j(k)$ can be obtained in (10), using the innovation sequence $\nu^j(k)$ and its covariance $S^j(k)$. The mode probability $p^j(k)$ corresponding to each candidate model at time index k is computed via (11). The initial mode probability is assumed to be $p^j(0) = 1/J$ without loss of generality. In the multiple-model approach, the identifiability conditions depend on each candidate model since the corresponding Kalman filters are non-interacting.

Identifiability conditions for estimating Q and R

Consider model j with the assumption of unknown Q^j and R^j . Now, consider the innovations corresponding to a stable, suboptimal closed-loop filter matrix $\bar{F}^j = F^j(I_{n_x} - W^j H^j)$ given by [4, 18]

$$\begin{aligned} \nu^j(k) = & H^j (\bar{F}^j)^m \tilde{x}^j(k-m|k-m-1) + \left\{ H^j \sum_{\ell=0}^{m-1} (\bar{F}^j)^{m-1-\ell} \right. \\ & \left. \times [\Gamma^j v^j(k-m+\ell) - F^j W^j w^j(k-m+\ell)] \right\} + w^j(k) \end{aligned} \quad (12)$$

where $\tilde{x}^j(k-m|k-m-1) = x^j(k-m) - \hat{x}^j(k-m|k-m-1)$ is the predicted error at time $(k-m)$. When the innovation sequence (12) is known, a weighted sum of innovations can be defined as

$$\xi^j(k) = \sum_{i=0}^m a_i^j \nu^j(k-i) \quad (13)$$

where the weights are the coefficients of the minimal polynomial of the closed-loop filter matrix $\bar{F}^j$, $\sum_{i=0}^m a_i^j (\bar{F}^j)^{m-i} = 0$, $a_0^j = 1$. It is easy to see that $\xi^j(k)$ is the sum of two moving average processes driven by the process noise and measurement noise, respectively, given by [4, 18]

$$\xi^j(k) = \sum_{l=1}^m B_l^j v^j(k-l) + \sum_{l=0}^m G_l^j w^j(k-l) \quad (14)$$

Here, B_l^j and G_l^j are given by

$$B_l^j = H^j \left(\sum_{i=0}^{l-1} a_i^j (\bar{F}^j)^{l-i-1} \right) \Gamma^j \quad (15)$$

$$G_l^j = \left[a_l^j I_{n_z} - H^j \left(\sum_{i=0}^{l-1} a_i^j (\bar{F}^j)^{l-i-1} \right) F^j W^j \right], G_0^j = I_{n_z} \quad (16)$$

Then, we obtain L_ℓ^j , the cross-covariance between $\xi^j(k)$ and $\xi^j(k-\ell)$, given by

$$\begin{aligned} L_\ell^j &= E[\xi^j(k) \xi^j(k-\ell)'] \\ &= \sum_{i=\ell+1}^m B_i^j Q^j (B_{i-\ell}^j)' + \sum_{i=\ell}^m G_i^j R (G_{i-\ell}^j)' \end{aligned} \quad (17)$$

The noise covariance matrices $Q^j = [q_{i\ell}]$ of dimension $n_v \times n_v$ and $R^j = [r_{i\ell}]$ of dimension $n_z \times n_z$ are positive definite and symmetric. By converting noise covariance matrices and the L_ℓ^j matrices as vectors, Zhang *et al.* [4] show that they are related by the noise covariance identifiability matrix I^j as in (18).

$$I^j \begin{bmatrix} \text{vec}(Q^j) \\ \text{vec}(R^j) \end{bmatrix} = \begin{bmatrix} L_0^j \\ L_1^j \\ \vdots \\ L_m^j \end{bmatrix} \quad (18)$$

As shown in [4], if matrix I^j has full rank, then the unknown noise covariance matrices, Q^j and R^j , are uniquely identifiable.

Recursive fading memory-based Innovation Correlation Estimation

We compute the sample correlation matrix $\hat{C}_{seq}^{j,k}(i)$ at sample k for model j and time lag i as a weighted combination of the correlation matrix $\hat{C}_{seq}^{j,k-1}(i)$ at the previous sample $(k-1)$ for model j and time lag i , and the samples of innovations $\nu^j(k-i)$ and $\nu^j(k)$. The tuning parameter λ , a positive constant between 0 and 1, is the weight associated with the previous sample correlation matrix. The recursive nature of the proposed algorithm makes it amenable to estimate Q^j and R^j for non-stationary systems.

The current M sample correlation matrices at time k are used as the initial values for the next pairs of samples for recursive computation. Let us define the number of samples as N .

$$\hat{C}_{seq}^{j,k}(i) = (1-\lambda) \nu^j(k-i) \nu^j(k)' + \lambda \hat{C}_{seq}^{j,k-1}(i) \quad (19)$$

$$\hat{C}_{seq}^{j,0}(i) = 0, i = 0, 1, 2, \dots, M-1; k = M, \dots, N \quad (20)$$

Objective function and the Gradient

The ensemble cross-correlations of a steady-state suboptimal Kalman filter is related to the closed-loop filter matrix $\bar{F}^j = F^j(I_{n_x} - W^j H^j)$, the matrix F^j , the measurement matrix H^j , the steady-state predicted covariance matrix $\bar{P}^j$, filter gain W^j and the innovation covariance, $C^j(0)$ via [8, 9]

$$\begin{aligned} C^j(i) &= E[\nu^j(k) \nu^j(k-i)'] \\ &= H^j (\bar{F}^j)^{i-1} F^j [\bar{P}^j (H^j)' - W^j C^j(0)] \end{aligned} \quad (21)$$

The objective function Ψ^j , formulated in [4], involves minimization of the sum of normalized $C^j(i)$ with respect to the corresponding diagonal elements of $C^j(0)$ for $i > 0$. Formally, we can define the objective function Ψ^j to be minimized with respect to W^j as

$$\begin{aligned} \Psi^j &= \frac{1}{2} \text{tr} \left\{ \sum_{i=1}^{M-1} [\text{diag}(C^j(0))]^{-\frac{1}{2}} C^j(i)' \right. \\ &\quad \left. \times [\text{diag}(C^j(0))]^{-1} C^j(i) [\text{diag}(C^j(0))]^{-\frac{1}{2}} \right\} \end{aligned} \quad (22)$$

where $\text{diag}(C^j)$ denotes the diagonal matrix of C^j or equivalently the Hadamard product of an identity matrix with C^j . We can rewrite the objective function by substituting (21) into (22) as

$$\Psi^j = \frac{1}{2} \text{tr} \left\{ \sum_{i=1}^{M-1} \phi^j(i) X^j \varphi^j (X^j)' \right\} \quad (23)$$

where

$$\phi^j(i) = [H^j (\bar{F}^j)^{i-1} F^j]' \varphi^j [H^j (\bar{F}^j)^{i-1} F^j] \quad (24)$$

$$X^j = \bar{P}^j (H^j)' - W^j C^j(0) \quad (25)$$

$$\varphi^j = [\text{diag}(C^j(0))]^{-1} \quad (26)$$

The gradient of objective function $\nabla_W \Psi^j$ can be computed as [4]

$$\nabla_W \Psi^j = - \sum_{i=1}^{M-1} [H^j (\bar{F}^j)^{i-1} F^j]' \varphi^j C^j(i) \varphi^j C^j(0) - (F^j)' Z^j F^j X^j - \sum_{l=0}^{i-2} [C^j(l+1) \varphi^j C^j(i)' \varphi^j H^j (\bar{F}^j)^{i-l-2}]' \quad (27)$$

The Z^j term in (27) is computed by the Lyapunov equation.

$$Z^j = (\bar{F}^j)' Z^j \bar{F}^j + \frac{1}{2} \sum_{i=1}^{M-1} (H^j (\bar{F}^j)^{i-1} F^j)' \varphi^j C^j(i) \varphi^j H^j + [(H^j (\bar{F}^j)^{i-1} F^j)' \varphi^j C^j(i) \varphi^j H^j]' \quad (28)$$

In computing the objective function and the gradient, we replace $C^j(i)$ by their sample estimates, $\hat{C}_{seq}^{j,k}(i)$. With this replacement, the noise covariance estimation becomes a data-dependent stochastic optimization/learning problem.

Updating gain W sequentially

Let B be the mini-batch size and let $K = N/B$ be the number of mini-batches. While the sequential mini-batch SGD estimation algorithm sequentially updates the M sample covariance matrices at every observation sample, we update the Kalman filter gain W when the sample index k is divisible by the size of mini-batch B . Our method allows more opportunities to converge to a better local minimum by frequently updating the gain when compared to the batch algorithm [5]. The generic form of the gain update is given by

$$(W^j)^{r+1} = (W^j)^r - (\alpha^j)^r (\nabla_W \Psi^j)^r \quad (29)$$

where r is the updating index starting with $r = 0$. Our previous research [5] presented the performance of incremental gradient methods (e.g., bold driver [19], constant, subgradient [20], RMSProp [21], Adam [22], Adadelata [23]) in the procedure of updating adaptive step size $\alpha^{(r)}$ in (29). In this paper, we applied the root mean square propagation (RMSProp) method. The RMSProp keeps track of the moving average of the squared incremental gradients for each gain element by adapting the step size element-wise.

$$\tau_{r,il}^j = \gamma^j \tau_{r-1,il}^j + (1 - \gamma^j) [(\nabla_W \Psi^j)_{il}^r]^2 \quad (30)$$

$$(\alpha_{il}^j)^r = \frac{(\alpha^j)^0}{\sqrt{\tau_{r,il}^j + \epsilon}}; \tau_0^j = 0; (\alpha^j)^0 = \frac{c}{K} \quad (31)$$

where $c > 0$ is a constant and K is the number of mini-batches. Here, $\gamma = 0.9$ is the default value and $\epsilon = 10^{-8}$ to prevent division by zero.

Estimation of R

Let $\mu^j(k)$ be the post-fit residual sequence of the Kalman filter, which is related to the innovations $\nu^j(k)$, $k = 1, 2, \dots, N$ via

$$\mu^j(k) = z(k) - H^j \hat{x}^j(k|k) = [I_{n_z} - H^j W^j(k)] \nu^j(k) \quad (32)$$

From the joint covariance of the innovation sequence $\nu^j(k)$ and the post-fit residual sequence $\mu^j(k)$, and the Schur determinant identity [24, 25], one can show that at the steady-state (assuming constant gain, W^j and constant Q^j and R^j over large enough time intervals such that the filter achieves steady state) [4]

$$G^j = E[\mu^j(k) \mu^j(k)'] = R^j (S^j)^{-1} R^j \quad (33)$$

where S^j is the steady-state innovation covariance. Because (33) can be interpreted as a simultaneous diagonalization problem in linear algebra [25] or as a continuous-time algebraic Riccati equation, the measurement covariance R^j can be estimated by solving the simultaneous diagonalization problem via Cholesky decomposition and eigen decomposition, or by solving a continuous-time Riccati equation as in [4, 26].

Estimation of Q

Given the estimated R^j , the process noise covariance Q^j and the steady-state updated state covariance P^j can be computed because these two parameters are generally coupled, requiring iterative computations [4]. Let t and l denote the iteration indices starting with $t = 0$ and $l = 0$. We can compute $(P^j)^0$, the initial steady-state updated covariance matrix, as the solution of the Lyapunov equation defined as

$$(P^j)^0 = \tilde{F}^j (P^j)^0 (\tilde{F}^j)' + W^j R^j (W^j)' + (I_{n_x} - W^j H^j) \Gamma^j (Q^j)^t (\Gamma^j)' (I_{n_x} - W^j H^j)' \quad (34)$$

where $\tilde{F}^j = (I_{n_x} - W^j H^j) F^j$ and $(Q^j)^0 = W^j S^j W^j'$ are the exact solution in the Wiener process case [4]. We iteratively update P^j as in (35) until convergence.

$$(P^j)^{t+1} = \left[(F^j (P^j)^t (F^j)' + \Gamma^j (Q^j)^t (\Gamma^j)' + (H^j)' (R^j)^{-1} H^j) \right]^{-1} \quad (35)$$

Given the converged P^j , Q^j will be updated in the t -loop until the estimate of Q^j converges.

$$(Q^j)^{t+1} = (\Gamma^j)^\dagger \left[(P^j + W^j S^j (W^j)' - F^j P^j (F^j)')^{t+1} \right] ((\Gamma^j)')^\dagger \quad (36)$$

The pseudocode for the sequential mini-batch SGD estimation algorithm for noise covariance parameters in the multiple-model approach is included as Algorithm 1.

Algorithm 1 Pseudocode of single-pass estimation algorithm in multiple-model adaptive Kalman filtering for a non-stationary system

```

1: input:  $(W^j)^0, (Q^j)^0, (R^j)^0, (\alpha^j)^0, B, p^j(0)$ 
   ▷  $(W^j)^0$ : initial gain,  $(Q^j)^0$ : initial  $Q^j$ ,  $(R^j)^0$ : initial
    $R^j$ ,  $(\alpha^j)^0$ : initial step size,  $B$ : batch size,  $p^j(0)$ : initial
   mode probability
2:  $r = 0$  {▷ Initialize the updating index  $r$ }
3: for  $k = 1$  to  $N$  do {▷  $N$ : Number of samples}
4:   for  $j = 1$  to  $J$  do {▷  $J$ : Number of models}
5:     compute innovation correlations  $\nu^j(k)$  as in (4)
6:     get the mode likelihood function  $\Lambda^j(k)$  by (10)
7:     if  $k > N_b + M$  then {▷  $N_b$ : Number of burn-in samples}
8:       compute  $\hat{C}_{seq}^{j,k}(i)$  as in (19)
9:       if  $\text{Mod}(k, B) = 0$  then
10:        obtain the objective function  $\Psi^j$  via (23)
11:        obtain the gradient  $\nabla_W \Psi^j$  via (27)
12:        update the step size  $(\alpha^j)^r$  via (31)
13:        update the gain via (29)
14:        ▷  $(W_{il}^j)^{r+1} = (W_{il}^j)^r - (\alpha_{il}^j)^r (\nabla_W \Psi^j)_{il}^r$ 
15:        get  $(R^j)^{r+1}$  by (33) and  $(Q^j)^{r+1}$  by (36)
16:         $r = r + 1$ 
17:      end if
18:    end for
19:    compute the mode probability  $p^j(k)$  via (11)
20:    mix the state estimates
21:  end for

```

3. NUMERICAL EXAMPLES

In this section, we present the utility of our sequential mini-batch SGD estimation algorithm in non-stationary systems. For the non-stationary systems, we assumed the process and measurement noise covariances continuously change. We first define the number of subgroups in which the noise covariances are not changing as N_{sg} . Given the number of observation samples, N , each subgroup has constant noise covariances with N/N_{sg} samples. Here, we applied the moving average algorithm to convert the piecewise constant noise covariances to continuous values. The fixed window slides down the piecewise constant noise covariances, computing the Gaussian-weighted moving average over each window size of N/N_{sg} . In this section, we consider two non-stationary scenarios for tracking time-varying Q and R with $N_{sg} = 5$ subgroups.

We also consider a scenario with an unknown motion model

using a multiple-model approach. The multiple-model algorithm estimates the noise covariance parameters using two Kalman filters, each tuned to a model, and the mode probabilities are used to infer which model is active. Here, we assume that a model is correct if its mode probability is greater than 0.66 and use the corresponding state estimate. We mix the state estimates by their mode probabilities if the mode probability of a model is between 0.33 and 0.66.

We present the performance of the sequential mini-batch SGD estimation algorithm using a single mode ($j = 1$), and then explore the multiple-model approach ($j = 1, 2$). Note that the number of “burn-in” samples and the number of lags are $N_b = 50$ and $M = 5$, respectively in the estimation procedure. The root mean square propagation (RMSProp) method is applied to update the filter gain. All Monte Carlo simulations were run using a computer with an Intel Core i7-8665U processor and 16 GB of RAM.

We used the averaged normalized innovation squared (NIS) metric [3] to measure the consistency of the proposed algorithm.

$$\bar{\varepsilon}_\nu(k) = \frac{1}{N_{mc}} \sum_{i=1}^{N_{mc}} \nu(k)' S^{-1} \nu(k) \quad (37)$$

where N_{mc} is the number of Monte Carlo runs.

Case 1: A detectable (but not completely observable) system

The incompletely observable (but detectable ²) system used in [7] is described by

$$F^1 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.2 \end{bmatrix}, H^1 = [1 \ 0], \Gamma^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (38)$$

The true values of Q and R are generated by first starting with piecewise constant functions for the five subgroups are generated based on $Q=[0.16, 0.49, 0.25, 0.36, 0.20]$, and $R = [0.30, 0.81, 0.49, 0.72, 0.42]$ with the values changing every 10,000 samples. The Gaussian-weighted moving average algorithm with a window size of 10,000 samples is applied to the piecewise constant functions to generate continuous noise covariances.

Table 1 shows the results of 100 Monte Carlo simulations based on the single-pass SGD algorithm in estimating Q and R . As can be seen, the estimated Q and R are close to their corresponding true values.

²The pair (F, H) in the system should be detectable in order for the continuous-time algebraic Riccati equation to have at least one positive semidefinite solution and in this case at least one solution results in a marginally stable steady-state KF [27, 28].

Table 1: Single-pass SGD estimation for case 1 (100 MC Runs; 50,000 samples; M = 5; RMSProp; Batch size = 64)

Subgroup index	R			Q			$\bar{P}_{11}$			$\bar{P}_{22}$			W_{11}			W_{21}		
	Truth	Mean	RMSE	Truth	Mean	RMSE	Truth	Mean	RMSE	Truth	Mean	RMSE	Truth	Mean	RMSE	Truth	Mean	RMSE
1 st	0.52	0.53	0.05	0.30	0.26	0.03	0.30	0.27	0.04	1.24	1.10	0.18	0.37	0.34	0.03	0.74	0.67	0.03
2 nd	0.64	0.68		0.36	0.33		0.36	0.33		1.48	1.35		0.36	0.34		0.73	0.67	
3 rd	0.60	0.59		0.30	0.29		0.30	0.29		1.24	1.20		0.34	0.34		0.68	0.67	
4 th	0.56	0.57		0.28	0.28		0.28	0.28		1.13	1.14		0.33	0.34		0.67	0.67	
5 th	0.42	0.41		0.20	0.20		0.20	0.20		0.83	0.83		0.33	0.33		0.66	0.67	

Figure 1 demonstrates that the sequential mini-batch gradient descent algorithm can track Q and R correctly. Here, the trajectories of Q and R estimates are smoothed by a simple first order fading memory filter with a smoothing weight of 0.7. Figure 1e shows the averaged NIS of SGD (RMSProp; batch size of 64) method with the 95% probability region [0.74, 1.30], and shows that the SGD-based Kalman filter is consistent.

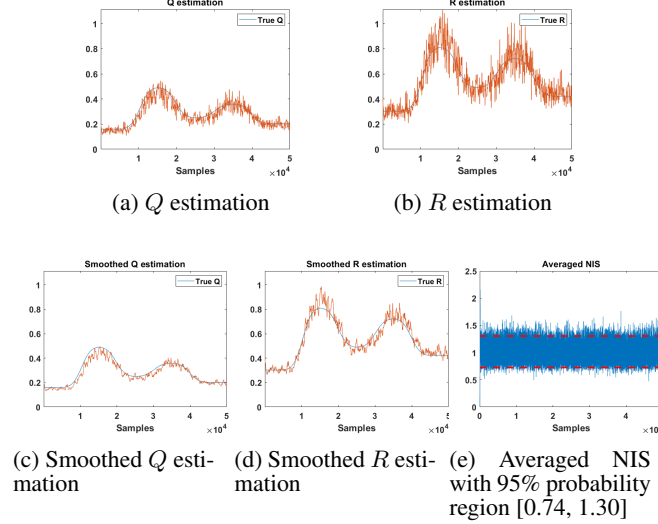


Figure 1: Trajectories of Q and R estimates without signal smoothing and with a smoothing weight of 0.7 for case 1

For the multiple-model approach, we assume that the measurements are generated by Model 1 as in (38). Model 2 is given by

$$F^2 = \begin{bmatrix} 0.8 & 1 \\ -0.6 & 0 \end{bmatrix}, H^2 = [1 \ 0], \Gamma^2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (39)$$

Figure 2 shows the trajectory of estimated parameters. The mode probability of first model is higher than the second model as expected. We find that the single-pass multiple-model approach can track Q and R accurately.

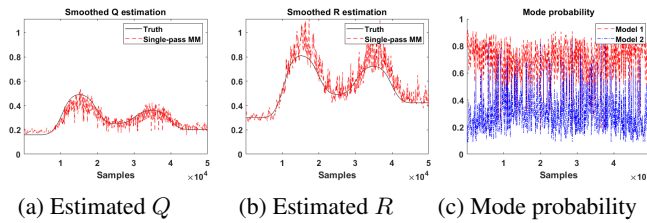


Figure 2: Trajectory of Q and R based on the multiple-model approach for case 1

Case 2: A five-state system with diagonal Q and R

The system matrices described in [8] are

$$F^1 = \begin{bmatrix} 0.75 & -1.74 & -0.3 & 0 & -0.15 \\ 0.09 & 0.91 & -0.0015 & 0 & -0.008 \\ 0 & 0 & 0.95 & 0 & 0 \\ 0 & 0 & 0 & 0.55 & 0 \\ 0 & 0 & 0 & 0 & 0.905 \end{bmatrix} \quad (40)$$

$$H^1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}; \Gamma^1 = \begin{bmatrix} 0 & 0 & 0 \\ 24.64 & 0 & 0 \\ 0 & 0.835 & 0 \\ 0 & 0 & 1.83 \end{bmatrix} \quad (41)$$

For this scenario, the piecewise values for Q and R are given in (42) and (43), and change every 20,000 samples. The continuous noise covariances are obtained by using Gaussian-weighted moving averaging with a window size of 20,000 samples.

$$\begin{bmatrix} Q_{11} \\ Q_{22} \\ Q_{33} \end{bmatrix} = \begin{bmatrix} [0.25, 0.64, 0.49, 0.25, 0.49] \\ [0.25, 0.36, 0.56, 0.16, 0.04] \\ [0.36, 0.49, 0.64, 0.25, 0.09] \end{bmatrix} \quad (42)$$

$$\begin{bmatrix} R_{11} \\ R_{22} \end{bmatrix} = \begin{bmatrix} [0.25, 0.56, 0.64, 0.42, 0.36] \\ [0.25, 0.25, 0.49, 0.16, 0.04] \end{bmatrix} \quad (43)$$

Table 2 shows the results of 100 Monte Carlo simulations for estimating the noise parameters using RMSProp update. The estimated parameters are close to the corresponding their true values. Given 100,000 samples, the proposed method with a batch-size of 64 requires 1,928 seconds for 100 Monte Carlo simulations, i.e., 193 seconds run.

Figure 3 shows the trajectories of the estimated Q and R with a signal smoothing with a smoothing weight of 0.7. For this example, it is known that accurate estimation of R_{11} is hard as shown in Figure 3d. The reason is that R_{11} is dominated by the state uncertainty, i.e., the measurement noise is “buried” in a much larger innovation [4]. In spite of the difficulty in estimating R_{11} , the filter is consistent as measured by NIS as shown in Figure 3f.

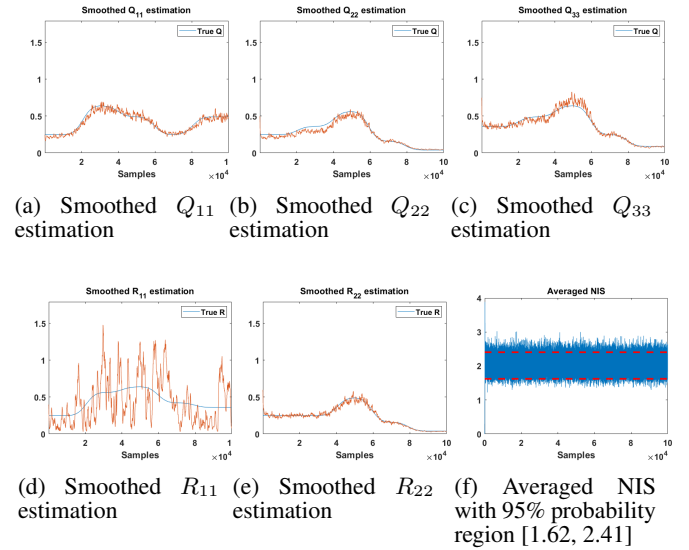


Figure 3: Trajectories of Q and R estimates with a signal smoothing at smoothing weight = 0.7 for case 2

Table 2: Single-pass SGD estimation for case 2 (100 MC Runs; 100,000 samples; $M = 5$; RMSProp; Batch size = 64)

(a) R , Q and $\bar{P}$ Estimates for case 2

Subgroup index	R_{11}		RMSE	R_{22}		RMSE	Q_{11}		RMSE	Q_{22}		RMSE	Q_{33}		RMSE
	Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean	
1 st	0.39	0.37	0.31	0.25	0.25	0.02	0.42	0.39	0.04	0.30	0.26	0.04	0.42	0.39	0.04
2 nd	0.60	0.64		0.36	0.34		0.56	0.57		0.46	0.39		0.56	0.57	
3 rd	0.53	0.70		0.30	0.31		0.36	0.37		0.33	0.33		0.42	0.45	
4 th	0.39	0.35		0.09	0.09		0.36	0.34		0.09	0.10		0.16	0.15	
5 th	0.36	0.30		0.04	0.04		0.49	0.48		0.04	0.04		0.09	0.08	
Subgroup index	$\bar{P}_{11}$		RMSE	$\bar{P}_{22}$		RMSE	$\bar{P}_{33}$		RMSE	$\bar{P}_{44}$		RMSE	$\bar{P}_{55}$		RMSE
	Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean	
1 st	30.22	28.10	3.79	0.38	0.35	0.05	508.64	473.07	69.97	0.28	0.25	0.05	4.71	4.35	0.54
2 nd	40.63	42.29		0.54	0.51		680.81	708.58		0.42	0.37		6.36	6.46	
3 rd	27.33	28.46		0.41	0.42		440.62	451.15		0.31	0.31		4.80	5.00	
4 th	23.26	22.04		0.13	0.13		422.27	406.27		0.09	0.09		1.77	1.67	
5 th	29.31	29.08		0.07	0.07		574.37	574.18		0.04	0.04		0.97	0.93	

(b) W Estimates for case 2

Subgroup index	W_{11}		RMSE	W_{21}		RMSE	W_{31}		RMSE	W_{41}		RMSE	W_{51}		RMSE
	Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean	
1 st	0.95	0.97	0.02	$3.09 \cdot 10^{-3}$	$-2.16 \cdot 10^{-3}$	0.01	-2.87	-2.76	0.07	$-5.37 \cdot 10^{-4}$	$6.30 \cdot 10^{-3}$	0.01	0.03	0.03	0.01
2 nd	0.95	0.97		$3.10 \cdot 10^{-3}$	$-4.62 \cdot 10^{-3}$		-2.86	-2.87		$-5.38 \cdot 10^{-4}$	$9.24 \cdot 10^{-3}$		0.03	0.01	
3 rd	0.94	0.95		$3.28 \cdot 10^{-3}$	$-2.93 \cdot 10^{-3}$		-2.80	-2.83		$-3.81 \cdot 10^{-4}$	$8.47 \cdot 10^{-3}$		0.04	0.03	
4 th	0.96	0.97		$3.80 \cdot 10^{-3}$	$1.75 \cdot 10^{-4}$		-2.95	-2.87		$-1.03 \cdot 10^{-3}$	$2.50 \cdot 10^{-3}$		0.02	0.02	
5 th	0.98	1.00		$3.50 \cdot 10^{-3}$	$7.28 \cdot 10^{-4}$		-3.03	-3.00		$-1.09 \cdot 10^{-3}$	$9.49 \cdot 10^{-4}$		0.01	0.01	
Subgroup index	W_{12}		RMSE	W_{22}		RMSE	W_{32}		RMSE	W_{42}		RMSE	W_{52}		RMSE
	Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean		Truth	Mean	
1 st	0.98	0.95	0.09	0.39	0.40	0.02	-1.64	-1.65	0.25	0.25	0.24	0.03	-0.98	-0.94	0.05
2 nd	0.93	0.99		0.38	0.41		-1.61	-1.71		0.27	0.24		-0.93	-0.97	
3 rd	0.91	0.96		0.37	0.38		-1.57	-1.68		0.25	0.25		-0.90	-0.93	
4 th	1.06	0.98		0.41	0.41		-1.49	-1.69		0.22	0.25		-1.05	-1.00	
5 th	1.19	1.04		0.44	0.46		-1.26	-1.75		0.20	0.23		-1.17	-1.11	

For the multiple-model approach, we consider two dynamic models. We generate measurement data based on Model 1 as in (40) and (41). Model 2 is given by

$$F^2 = \begin{bmatrix} 0.8 & -1.71 & -0.31 & 0 & -0.15 \\ 0.07 & 0.9 & -0.001 & 0 & -0.008 \\ 0 & 0 & 0.95 & 0 & 0 \\ 0 & 0 & 0 & 0.55 & 0 \\ 0 & 0 & 0 & 0 & 0.905 \end{bmatrix} \quad (44)$$

$$H^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}; \Gamma^2 = \begin{bmatrix} 0 & 0 & 0 \\ 24.64 & 0 & 0 \\ 0 & 0.835 & 0 \\ 0 & 0 & 1.83 \end{bmatrix} \quad (45)$$

Figure 4 shows the trajectory of estimated parameters. The first model shows higher mode probability compared to the second model. In terms of noise covariance estimation, the single-pass SGD estimation with a multiple-model algorithm can track Q and R .

4. CONCLUSION

In this paper, we presented a single-pass stochastic gradient descent (SGD) algorithm for the noise covariance estimation

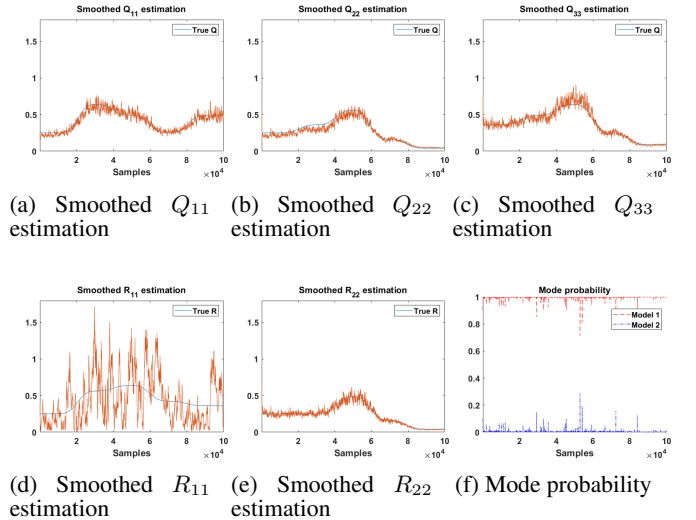


Figure 4: Trajectory of Q and R based on the multiple-model approach for case 2

in adaptive Kalman filters that has acceptable state estimation root mean square error (RMSE). This algorithm is valid for streaming data from non-stationary systems, where the noise covariances continuously change, as well as for multiple models. The evaluation of the proposed method on a number

of test cases demonstrated its accuracy and filter consistency over extant approaches. In the future, a number of research avenues can be pursued, including 1) estimating Q and R using one-step lag smoothed residuals; 2) automatic model selection from a library of dynamic models for model adaptation; and 3) exploration of the utility of the covariance estimation algorithm as an alternative to interacting multiple models (IMMs).

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REFERENCES

- [1] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 03 1960. [Online]. Available: 10.1115/1.3662552
- [2] F. Auger, M. Hilairet, J. M. Guerrero, E. Monmasson, T. Orlowska-Kowalska, and S. Katsura, "Industrial applications of the kalman filter: A review," *IEEE Transactions on Industrial Electronics*, vol. 60, no. 12, pp. 5458–5471, 2013. [Online]. Available: 10.1109/TIE.2012.2236994
- [3] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with applications to tracking and navigation: theory algorithms and software*. John Wiley & Sons, 2004.
- [4] L. Zhang, D. Sidoti, A. Bienkowski, K. R. Pattipati, Y. Bar-Shalom, and D. L. Kleinman, "On the identification of noise covariances and adaptive kalman filtering: A new look at a 50 year-old problem," *IEEE Access*, vol. 8, pp. 59 362–59 388, 2020. [Online]. Available: 10.1109/ACCESS.2020.2982407
- [5] H.-S. Kim, L. Zhang, A. Bienkowski, and K. R. Pattipati, "Multi-pass sequential mini-batch stochastic gradient descent algorithms for noise covariance estimation in adaptive kalman filtering," *IEEE Access*, vol. 9, pp. 99 220–99 234, 2021. [Online]. Available: 10.1109/ACCESS.2021.3094963
- [6] C. Neethling and P. Young, "Comments on" identification of optimum filter steady-state gain for systems with unknown noise covariances," *IEEE Transactions on Automatic Control*, vol. 19, no. 5, pp. 623–625, 1974. [Online]. Available: 10.1109/TAC.1974.1100694
- [7] B. J. Odelson, M. R. Rajamani, and J. B. Rawlings, "A new autocovariance least-squares method for estimating noise covariances," *Automatica*, vol. 42, no. 2, pp. 303–308, 2006. [Online]. Available: 10.1016/j.automatica.2005.09.006
- [8] R. Mehra, "On the identification of variances and adaptive kalman filtering," *IEEE Transactions on automatic control*, vol. 15, no. 2, pp. 175–184, 1970. [Online]. Available: 10.1109/TAC.1970.1099422
- [9] P. R. Belanger, "Estimation of noise covariance matrices for a linear time-varying stochastic process," *Automatica*, vol. 10, no. 3, pp. 267–275, 1974. [Online]. Available: 10.1016/0005-1098(74)90037-5
- [10] S. Sarkka and A. Nummenmaa, "Recursive noise adaptive kalman filtering by variational bayesian approximations," *IEEE Transactions on Automatic control*, vol. 54, no. 3, pp. 596–600, 2009. [Online]. Available: 10.1109/TAC.2008.2008348
- [11] R. Killick, P. Fearnhead, and I. A. Eckley, "Optimal detection of changepoints with a linear computational cost," *Journal of the American Statistical Association*, vol. 107, no. 500, pp. 1590–1598, 2012. [Online]. Available: 10.1080/01621459.2012.737745
- [12] H.-S. Kim, A. Bienkowski, and K. R. Pattipati, "A single-pass noise covariance estimation algorithm in multiple-model adaptive kalman filtering for non-stationary systems," *TechRxiv preprint:10.36227/techrxiv.14761005.v2*, 2022.
- [13] H.-S. Kim, L. Zhang, A. Bienkowski, and K. R. Pattipati, "A single-pass noise covariance estimation algorithm in adaptive kalman filtering for non-stationary systems," *2021 IEEE 24th International Conference on Information Fusion (FUSION)*, pp. 556–563, 2021. [Online]. Available: 10.23919/FUSION49465.2021.9626861
- [14] X.-R. Li and Y. Bar-Shalom, "Multiple-model estimation with variable structure," *IEEE Transactions on Automatic Control*, vol. 41, no. 4, pp. 478–493, 1996. [Online]. Available: 10.1109/9.48927
- [15] P. Hanlon and P. Maybeck, "Multiple-model adaptive estimation using a residual correlation kalman filter bank," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 36, no. 2, pp. 393–406, 2000. [Online]. Available: 10.1109/7.845216
- [16] S. Servadio and R. Zanetti, "Differential algebra-based multiple gaussian particle filter for orbit determination," *Journal of Optimization Theory and Applications*, vol. 191, no. 2, pp. 459–485, 2021. [Online]. Available: 10.1007/s10957-021-01934-8
- [17] R. S. Bucy and P. D. Joseph, *Filtering for stochastic processes with applications to guidance*. American Mathematical Soc., 2005, vol. 326.
- [18] K. Tajima, "Estimation of steady-state kalman filter gain," *IEEE Transactions on Automatic Control*, vol. 23, no. 5, pp. 944–945, 1978. [Online]. Available: 10.1109/TAC.1978.1101838
- [19] R. Battiti, "Accelerated backpropagation learning: Two optimization methods," *Complex systems*, vol. 3, no. 4, pp. 331–342, 1989.
- [20] S. Boyd, L. Xiao, and A. Mutapcic, "Subgradient methods," *lecture notes of EE392o, Stanford University, Autumn Quarter*, vol. 2004, pp. 2004–2005, 2003.
- [21] T. Tieleman and G. Hinton, "Lecture 6.5-rmsprop: Divide the gradient by a running average of its recent magnitude," *COURSERA: Neural networks for machine learning*, vol. 4, no. 2, pp. 26–31, 2012.
- [22] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv*, 2014. [Online]. Available: 10.48550/arXiv.1412.6980
- [23] M. D. Zeiler, "Adadelta: an adaptive learning rate method," *arXiv*, 2012. [Online]. Available: 10.48550/arXiv.1212.5701
- [24] D. Bertsekas, *Nonlinear Programming*, ser. Athena scientific optimization and computation series. Athena Scientific, 2016.

- [25] G. H. Golub and C. F. Van Loan, *Matrix computations*. JHU press, 2013, vol. 3.
- [26] W. Arnold and A. Laub, "Generalized eigenproblem algorithms and software for algebraic riccati equations," *Proceedings of the IEEE*, vol. 72, no. 12, pp. 1746–1754, 1984. [Online]. Available: 10.1109/PROC.1984.13083
- [27] F. Naets, J. Cuadrado, and W. Desmet, "Stable force identification in structural dynamics using kalman filtering and dummy-measurements," *Mechanical Systems and Signal Processing*, vol. 50, pp. 235–248, 2015.
- [28] D. Simon, *Optimal state estimation: Kalman, H infinity, and nonlinear approaches*. John Wiley & Sons, 2006.

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